

DOCUMENTATION

GROUP TRAVEL PLANNER PROJECT

Group Travel Planner:

Author: Abdul Manan

General Description

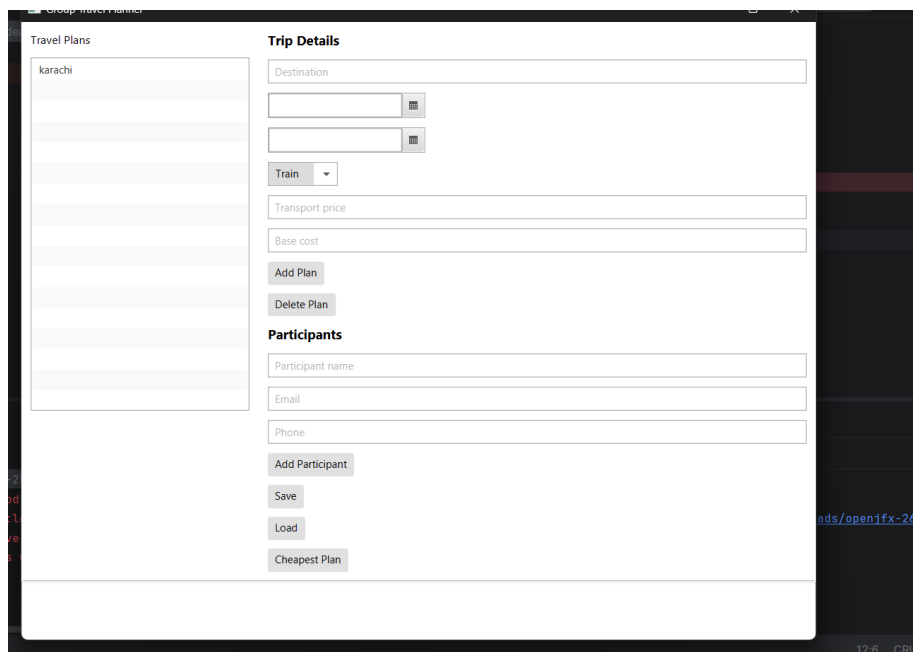
The Group Travel Planner is a small Java application which help a group of people to organize their trips in easy and structured way. The main purpose of this software is to make travel planning simple for users who want to compare different trip plans before making final decision.

This application is useful for students, friends, families, or small groups who are planning travel together. The users can add destination, travel dates, transport type, transport price, total base cost and participants details. The system also give option to save the travel plans in file and load them later when the application starts again.

The software is made by using JavaFX graphical user interface and object-oriented programming concepts. The application use multiple classes to separate the responsibilities such as trip data, participant data, transport option and file handling. This makes the project easy to understand and maintain.

2. Screenshots of the UI:

Main JavaFX UI



Running Application

Java Classes Used in the Project

The project uses several Java classes based on object-oriented programming.

1. TravelPlan

This class store the main trip information such as:

- destination
- start date
- end date
- transport type
- transport price
- base cost
- participants

This is the main class of the project.

2. Participant

This class store the participant information such as:

- name
- email
- phone number

This helps to manage all group members joining the trip.

3. Person

This class is parent class which store common personal details. Participant class use inheritance from this class.

4. TransportOption

This class represent different transport methods such as bus, train and flight.

5. TravelManager

This class handle the business logic of the application. It manage adding plans, deleting plans and storing trip list.

6. FileHandler

This class is responsible for file operations. It save and load travel plan data using Java serialization.

7. MainApp

This is the main JavaFX application class that launch the program window.

8. TravelPlannerUI

This class build the complete graphical interface using JavaFX controls such as buttons, list view, text fields and date pickers.

4. Brief Story of Project Evolution

At the beginning, the project idea was selected because travel planning is common activity for students and friends. I wanted to make something simple but practical that can be used in real life. First, I designed the class structure by thinking what kind of data a travel planner needs such as destination, dates, costs and participants.

After class planning, I started building the JavaFX user interface. First version only included destination and dates, but later I improved it by adding transport options, cost fields, participant management and file saving features. One challenge was connecting UI fields with the backend classes and making sure the selected trip is shown correctly in the form.

Another important part of project evolution was implementing file serialization. This allowed the software to save all travel plans in travelplans.dat file and restore them later. After that, I improved the layout to make it cleaner and easier for novice users.

5. Usage Instructions for a Novice User

The user first opens the application and enters the trip details including destination, travel start date, end date, transport type, transport price and base cost. After that, the user clicks **Add Plan** button to create the trip. The created plans appear in the left side list view.

To add participants, the user selects the trip, enters participant name, email and phone, then clicks **Add Participant**. The trip can be saved permanently by clicking **Save** button. Later, the user can reopen the program and click **Load** to restore all previous saved travel plans.

6. Self-Evaluation of the Project

The positive aspect of this project is that it successfully combines JavaFX GUI, object-oriented design and file handling in one simple software. The application is easy to use and fulfills all main course requirements such as multiple classes, inheritance, serialization and event handling.

One challenge was designing the user interface layout in clean way and connecting the data with list view updates. Another challenge was handling file saving correctly without data loss. However, after testing, the program works properly and provides stable functionality.

7. Further Development

In future, this project can be improved by adding trip comparison feature where users can compare multiple travel plans based on total cost and transport type. Another useful feature can be automatic cost split between participants.

The system can also be improved by adding weather API, hotel booking options and map integration. Better table-based UI design can also make the software more professional. In future version, database storage can replace simple serialization file to support larger amount of travel plans.